INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/03

Paper 3 Applications Paper

17 September 2015

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	15
2	15
3	10
4	12
5	20
Total	72

This document consists of **15** printed pages and **1** blank page.

Section A

Answer **all** questions.

1 *Agrobacterium tumefaciens* is a soil bacterium commonly used in the genetic engineering of plants.

Fig. 1.1 shows the stages in the transfer of the *cry* gene from *Bacillus thuringiensis* into plant cells.

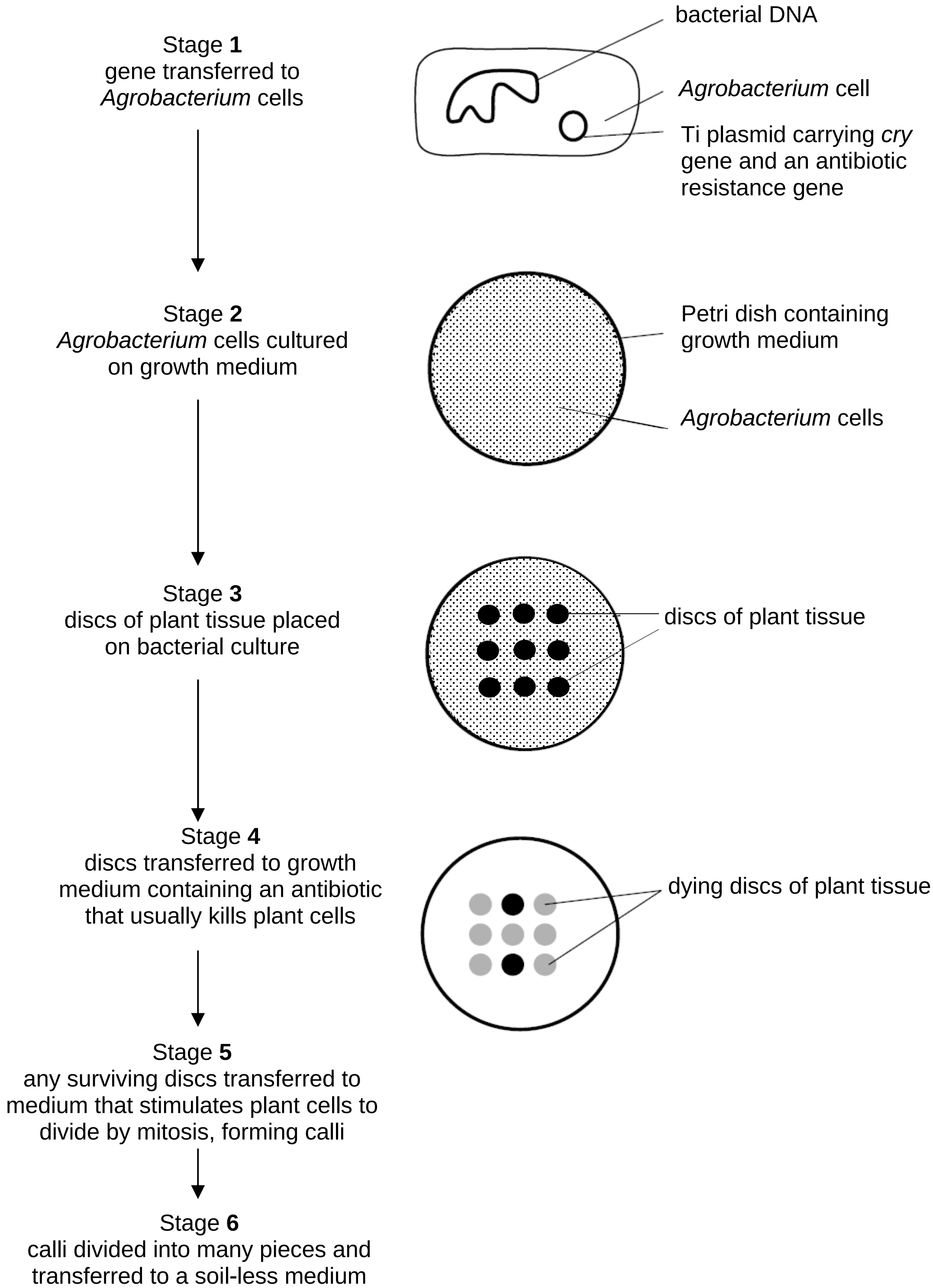

Fig. 1.1

(a) The *cry* gene can be obtained by screening DNA libraries.

With reference to the differences between genomic and cDNA libraries, explain which DNA library is preferred for cloning of the *cry* gene into plant cells.

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.... [2]

(b) Describe how the Ti plasmid containing the *cry* gene is produced.

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.... [2]

(c) Describe how the Ti plasmid is transferred into *Agrobacterium* cells.

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.... [2]

(d) With reference to Fig. 1.1,

(i) Explain the significance of the antibiotic resistance gene on the Ti plasmid.

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.... [2]

(ii) Suggest the relative concentrations of plant growth regulators present in the nutrient medium in Stage **5**.

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.... [1]

The fall armyworm is a common insect pest of field corn. The use of spray insecticides is ineffective as the larvae are protected once they enter the stem.

Two types of Bt corn, YieldGard and Herculex, have been genetically modified with variants of the *cry* gene coding for different forms of Bt toxin.

The performances of YieldGard and Herculex technologies for their effects on fall armyworm survival and corn damage were compared with a conventional non-Bt corn line. The results of the study are shown in Fig. 1.2.

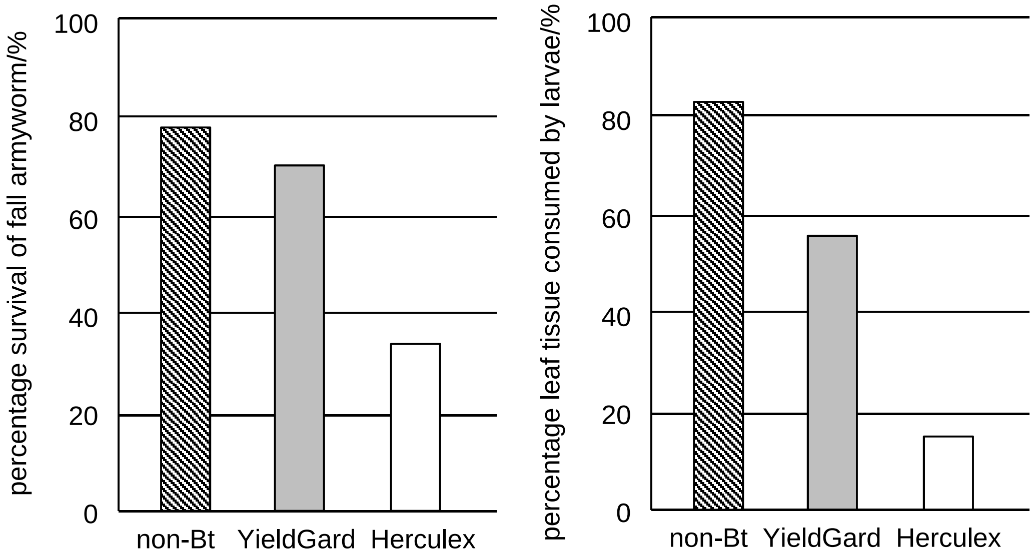

Fig. 1.2

(e) With reference to Fig. 1.2,

(i) Describe the effect of Bt toxin on the fall armyworm.

-- [2]

(ii) Deduce which type of Bt corn will best increase crop yield.

-- [2]

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(f) Describe **two** concerns of genetically modified Bt corn.

[2]

[Total: 15]

2 A cybrid (cytoplasmic hybrid cell) is produced as shown in Fig. 2.1.

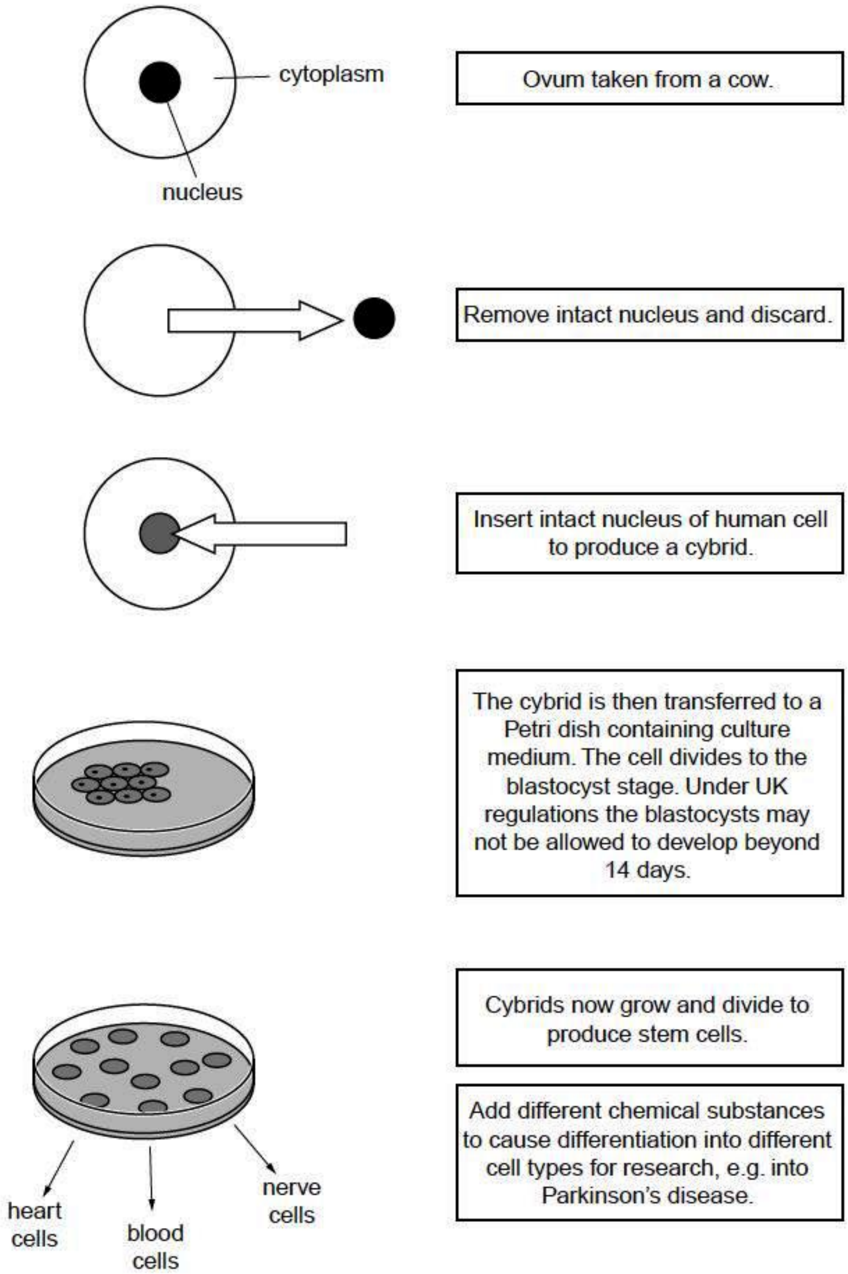

Fig. 2.1

(a) The DNA of such a cybrid is 99.6% human. The remaining 0.4% of the DNA is from cow in the cytoplasm. Explain why there is DNA in the cytoplasm of the cybrid.

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.... [2]

(b) Suggest why UK regulations do not allow blastocysts to develop beyond 14 days.

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.... [1]

(c) **(i)** State which cells are taken from the blastocyst to produce stem cells.

.... [1]

(ii) Describe the features of these stem cells.

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.... [3]

(d) **(i)** Suggest how addition of different chemical substances can cause stem cells to differentiate into different cell types.

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.... [2]

(ii) Explain if the heart cells, blood cells and nerve cells formed are human or bovine (cow).

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.... [1]

(iii) Suggest how the resultant cells can be used for research, e.g. Parkinson's disease.

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.... [2]

(e) When the Human Fertilisation and Embryology Bill was considered by the UK Parliament in 2008, some people argued that it is unethical to allow the production of cybrids.

State whether you **agree** or **disagree** that this is unethical and explain why you reached this decision.

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.... [3]

[Total: 15]

3 Cystic fibrosis (CF) affects epithelial cells lining organs such as the lungs and intestines leading to accumulation of mucus. In 1989, the mutated gene responsible for cystic fibrosis was identified. This opened the way for the development of gene therapy to reduce the symptoms of the disease. The idea has been to introduce a normal copy of the gene into the cells of patients.

(a) **(i)** Describe **one** mutation that leads to CF.

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.... [2]

(ii) Explain how the mutation stated in **(a)(i)** results in mucus accumulation in lung epithelial.

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.... [3]

Many attempts have been made to find methods of using gene therapy to treat cystic fibrosis. One approach uses viruses to deliver normal alleles of the CFTR gene into epithelial cells of the airways. Viral delivery systems have two main problems:

- The virus may trigger an immune response which destroys the infected cells.
- Most non-pathogenic viruses are not very good at getting into cells, so very few cells receive the allele.

A team of researchers in the USA developed a new strain of adeno-associated virus, AAV2.5T. Genes for the CFTR protein and for an enzyme, luciferase, were added to the DNA of the viruses. Luciferase produces a fluorescent green protein when luciferin is added.

The normal AAV strain and the AAV2.5T strain were added to cultures of epithelial cells from the airways. After adding luciferin, the numbers of cells that had taken up the CFTR gene was estimated using the intensity of the green fluorescence which developed.

The results are shown in Fig. 3.1.

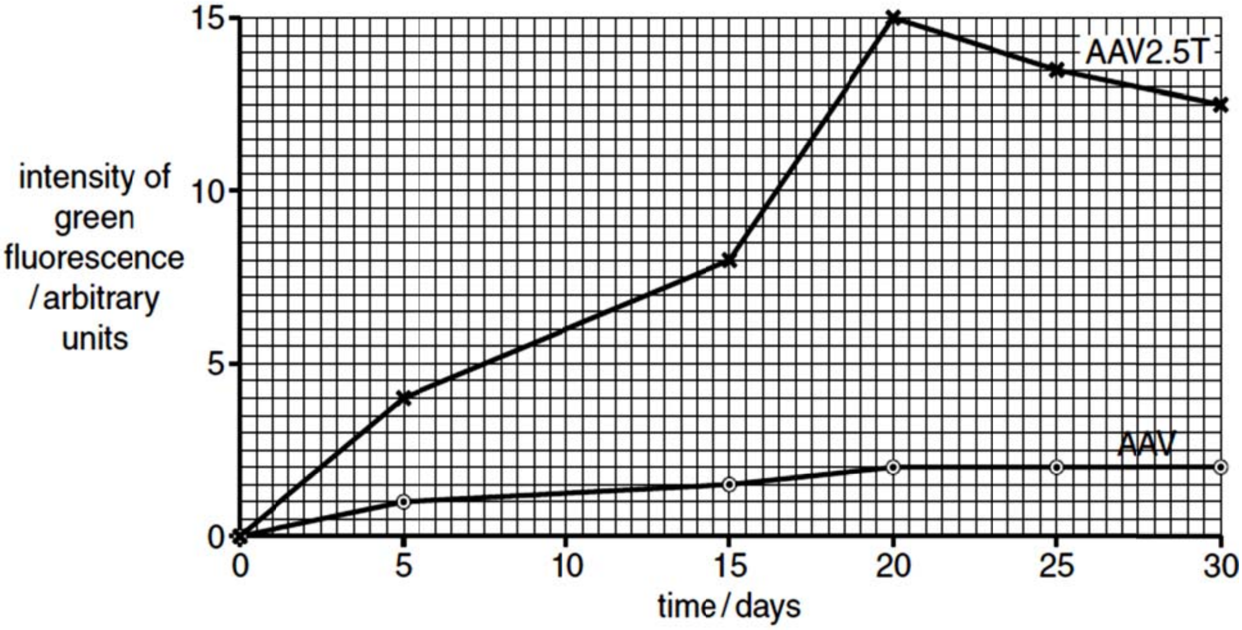

Fig. 3.1

(b) Explain why the intensity of green fluorescence can be used to estimate the number of cells that have taken up the CFTR gene.

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.... [2]

(c) With reference to Fig. 3.1, explain the difference in ability of the two viral strains, AAV and AAV2.5T, to infect epithelial cells from the airways.

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.... [2]

(d) State **one** other advantage of using AAV to deliver genes into cells.

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.... [1]

[Total: 10]

4 Planning question

Yeast is a single cell eukaryotic fungus used in industry for the production of drinks containing ethanol. The yeast is grown with a carbohydrate energy source in anaerobic conditions. During this fermentation process the yeast oxidises the carbohydrate to synthesise ATP for growth and produces ethanol and carbon dioxide as by-products.

The equation for anaerobic respiration is:

$$C_6H_{12}O_6 \longrightarrow 2C_2H_5OH + 2CO_2$$

You are required to find out the rate of respiration of yeast using different carbohydrates.

You are provided with the following materials and apparatus which you must use:

- activated yeast culture
- 10% fructose
- 10% glucose
- 10% lactose
- 10ml syringes
- boiling tubes
- thermostatically regulated water bath and thermometer

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated
- pipettes, glass rods etc.
- stopwatch

Your plan should:
- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s), if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

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Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a), (b),** etc., as indicated in the question.

5 (a) Discuss the problems associated with cloning of eukaryotic genes using *E. coli* cells. [5]

(b) Explain how problems associated with cloning of eukaryotic genes using *E. coli* cells are overcome. [6]

(c) Rubisco is an important enzyme in the Calvin cycle.

Outline how it might be possible to produce genetically modified plants with increased rubisco concentrations, or rubisco with enhanced efficiency. [9]

[Total: 20]

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